On Fixed Point Theorems Satisfying Compatibility Property in Modular G-Metric Spaces

Chong Han & James Evans

03/06/2026

Preliminaries

We begin this section by recalling some definitions and results which will be useful in this paper.

Theorem 2.1. ([9]) *Let* $\{a_n\}_{n\in\mathbb{N}}, \{b_n\}_{n\in\mathbb{N}}, \{c_n\}_{n\in\mathbb{N}}$ *be three sequences in* $\mathbb{R}$ *such that*

(i) $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = \ell$,

(ii) *for some positive integer* N*,* $a_n \le c_n \le b_n$ *for all* $n \ge N$*.*

Then $\lim\limits_{n\to\infty} c_n = \ell$*.*

Definition 2.2. ([14]) Self-mappings f and g of a metric space (X, d) are compatible if $\lim\limits_{n\to\infty} d(gf(x_n), fg(x_n)) = 0$, whenever $\{x_n\}_{n\ge 1}$ is a sequence in X such that $\lim\limits_{n\to\infty} f(x_n) = \lim\limits_{n\to\infty} g(x_n) = t$, for some t in X.

Definition 2.3. ([15]) Let f and g be mappings from a G-metric space (X, G) into itself. The mappings f and g are said to be compatible if there exists a sequence $\{x_n\}$ such that

$$\lim_{n\to\infty} G(fgx_n, gfx_n, gfx_n) = 0$$

The logical gap. From elementary calculus one learns:

$$x_n \to t \;\Rightarrow\; f(x_n) \to f(t).$$

In analysis, this statement is in fact an *if and only if* characterization of continuity at the point t. Thus, from

$$f(x_n) \to t,$$

we are **not** permitted to conclude

$$g(f(x_n)) \to g(t),$$

since continuity of g at t has not been assumed.

Application to the definition. We know

$$f(x_n) \to t \quad \text{and} \quad g(x_n) \to t,$$

so both sequences $f(x_n)$ and $g(x_n)$ approach the same point t. Nevertheless, the composed sequences $g(f(x_n))$ and $f(g(x_n))$ are not determined by $g(t)$ and $f(t)$ without continuity assumptions. Compatibility imposes

$$d\big(g(f(x_n)),\, f(g(x_n))\big) \to 0.$$

Hence, even though neither limit can be evaluated individually, the two compositions must converge to the **same** limit (if they converge).

Purpose of the definition. Compatibility serves as a replacement for the missing continuity step. If g were continuous at t, we could write

$$\lim_{n\to\infty} g(f(x_n)) \;=\; g\Big(\lim_{n\to\infty} f(x_n)\Big) \;=\; g(t).$$

Since continuity is not assumed, compatibility is imposed instead to ensure that the two compositions asymptotically coincide.

Definition 2.4 (Modular G-metric; inserted for completeness). Let X be a nonempty set. A *modular G-metric* on X is a family of functions

$$\omega^G_\lambda : X \times X \times X \to [0,\infty], \qquad \lambda > 0,$$

satisfying, for all $x, y, z, a^* \in X$ and all $\lambda, \mu > 0$:

1. $\omega^G_\lambda(x,y,z) = 0$ if $x = y = z$.
2. $\omega^G_\lambda(x,x,y) > 0$ if $x \neq y$.
3. $\omega^G_\lambda(x,x,y) \leq \omega^G_\lambda(x,y,z)$ if $z \neq y$.
4. $\omega^G_\lambda(x,y,z)$ is symmetric in (x,y,z).
5. (**Rectangle inequality**)

$$\omega^G_{\lambda+\mu}(x,y,z) \leq \omega^G_\lambda(x,a^*,a^*) + \omega^G_\mu(a^*,y,z).$$

The pair (X, ω^G) is then called a *modular G-metric space*.

Intuition: why "$\leq$" rather than equality (midpoint intuition) Unlike a metric $d(x,y)$, which is fundamentally a 2-point notion with triangle-inequality chaining, $\omega^G_\lambda(x,y,z)$ is a *3-point collapse/spread* quantity: it measures how far the triple (x,y,z) is from "collapsing." If one tries to split (x,y,z) through an intermediate point a^*, information is unavoidably lost (especially about the direct y–z interaction). Thus only an *upper bound* is natural and sufficient for proof chaining. The rectangle inequality replaces triangle-inequality telescoping.

Definition 2.5. ([5]) Let $(X_\omega, \omega_\lambda^G)$ be a modular G-metric space. The sequence $\{x_n\}_{n\in\mathbb{N}}$ in X_{ω^G} is modular G-convergent to x^*, if it converges to x^* in the topology $\tau(\omega_\lambda^G)$.

A function $T : X_\omega \to X_\omega$ at $x^* \in X_{\omega^G}$ is called modular G-continuous if $\omega_\lambda^G(x_n, x^*, x^*) \to 0$ then $\omega_\lambda^G(Tx_n, Tx^*, Tx^*) \to 0$, for all $\lambda > 0$. The sequence $\{x_n\}_{n\in\mathbb{N}}$ modular G-converges to x^* as $n \to \infty$, if $\lim\limits_{n\to\infty} \omega_\lambda^G(x_n, x_m, x^*) = 0$. That is for all $\epsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $\omega_\lambda^G(x_n, x_m, x^*) < \epsilon$ for all $n, m \geq n_0$. Here we say that x^* is modular G-limit of $\{x_n\}_{n\in\mathbb{N}}$.

Without any confusion we will take X_{ω^G} as a modular ω^G-metric space.

Definition 2.6. ([5]) Let (X_ω, ω^G) be a modular ω^G-metric space. Then $\{x_n\} \subseteq X_{\omega^G}$ is said to be modular ω^G-Cauchy if for every $\epsilon > 0$, there exists $n_\epsilon \in \mathbb{N}$ such that $\omega_\lambda^G(x_n, x_m, x_l) < \epsilon$ for all $n, m, l \geq n_\epsilon$ and $\lambda > 0$.

A modular G-metric space X_{ω^G} is said to be modular G-complete if every modular ω^G-Cauchy sequence in X_{ω^G} is modular ω^G-convergent in X_{ω^G}.

Definition 2.5 (Convergence). A sequence $\{x_n\}$ is modular G-convergent to x^* if

$$\omega_\lambda^G(x_n, x^*, x^*) \to 0 \qquad \text{(for every } \lambda > 0\text{)}.$$

Interpretation. In an ordinary metric space,

$$x_n \to x^* \iff d(x_n, x^*) \to 0.$$

Here there is no two-point distance $d(x, y)$; instead one measures a three-point spread. Thus, to test whether x_n approaches x^*, one asks whether the triple (x_n, x^*, x^*) collapses to a single point. In this sense, convergence means that the configuration shrinks onto x^*.

Equivalent form. They also express this as

$$\lim_{n,m\to\infty} \omega_\lambda^G(x_n, x_m, x^*) = 0,$$

which says that all sufficiently late terms of the sequence cluster around x^* simultaneously. This is the three-point analogue of requiring $|x_n - x^*| \to 0$.

Epsilon formulation. Equivalently, for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that

$$\omega_\lambda^G(x_n, x_m, x^*) < \varepsilon \qquad \text{for all } n, m \geq n_0.$$

In this case x^* is called the modular G-limit of $\{x_n\}$.

Modular G-continuity. A mapping T is modular G-continuous at x^* if

$$\omega_\lambda^G(x_n, x^*, x^*) \to 0 \quad \Rightarrow \quad \omega_\lambda^G(Tx_n, Tx^*, Tx^*) \to 0 \qquad \text{(for every } \lambda > 0\text{)}.$$

This is exactly the usual implication $x_n \to x^* \Rightarrow Tx_n \to Tx^*$, rewritten in the modular G-metric.

Definition 2.6 (Cauchy sequence). The sequence $\{x_n\}$ is modular ω^G-Cauchy if for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that

$$\omega_\lambda^G(x_n, x_m, x_\ell) < \varepsilon \qquad \text{for all } n, m, \ell \geq n_\varepsilon \text{ and all } \lambda > 0.$$

This is the analogue of the usual condition $d(x_n, x_m) \to 0$, but expressed in terms of three-point collapse.

Completeness. A modular G-metric space is modular G-complete if every modular ω^G-Cauchy sequence is modular ω^G-convergent.

One-sentence intuition. A sequence converges when every sufficiently late portion of it becomes indistinguishable from a single point, where indistinguishability is measured by three-point collapse rather than by pairwise distance.

Proposition 2.7. ([5]) *Let (X_ω, ω^G) be a modular ω^G-metric space, for any $x, y, z, a \in X_{\omega^G}$, it follows that:*

(1) *If $\omega_\lambda^G(x, y, z) = 0$ for all $\lambda > 0$, then $x = y = z$.*
(2) $\omega_\lambda^G(x, y, z) \leq \omega_{\frac{\lambda}{2}}^G(x, x, y) + \omega_{\frac{\lambda}{2}}^G(x, x, z)$ *for all* $\lambda > 0$.
(3) $\omega_\lambda^G(x, y, y) \leq 2\omega_{\frac{\lambda}{2}}^G(y, x, x)$ *for all* $\lambda > 0$.
(4) $\omega_\lambda^G(x, y, z) \leq \omega_{\frac{\lambda}{2}}^G(x, a, z) + \omega_{\frac{\lambda}{2}}^G(a, y, z)$ *for all* $\lambda > 0$.

The midpoint intuition. One might imagine the Euclidean picture:

$$d(y, z) = d(y, m) + d(m, z)$$

when m lies on the segment between y and z (for example, a midpoint). The question is why a similar equality does not hold for

$$\omega_\lambda^G(x, y, z).$$

Metric versus G-metric measurement. In a metric space,

$$d(y, z)$$

measures a pairwise separation. It admits a path interpretation: one can think of moving from y to z, and decomposition through an intermediate point corresponds to splitting that path. In contrast,

$$\omega_\lambda^G(x, y, z)$$

measures a simultaneous incompatibility of three points. It answers how far the triple (x, y, z) is from collapsing to a single point. There is no notion of segment, between-ness, or midpoint in this structure, and therefore no path interpretation.

Why splitting loses information. The triple (x, y, z) encodes three relationships: the relation of y to x, the relation of z to x, and the relation of y to z. If we replace it by

$$(x, x, y) \quad \text{and} \quad (x, x, z),$$

we retain only the relations to x and discard the direct interaction between y and z. Consequently, these pieces cannot reconstruct the original configuration; they only provide an upper bound, which is why the axiom yields

$$\leq$$

rather than equality.

Key idea. In a metric space, distance behaves like path length and can sometimes be decomposed exactly. In a modular G-metric space, the quantity measures collective disagreement of three points, so decomposition gives only an estimate.

Compatibility Notions

Classical compatibility (metric/G-metric)

In classical fixed point theory:

- In a metric space (X, d), maps f, g are *compatible* if $d(gf(x_n), fg(x_n)) \to 0$ whenever $f(x_n) \to t$ and $g(x_n) \to t$.
- In a G-metric space (X, G), the definition is adapted using $G(\cdot, \cdot, \cdot)$, and appears in two equivalent symmetric forms.

These notions formalize "asymptotic commutation" on sequences converging to the same point.

ω-compatibility in modular G-metric spaces

The paper introduces a modular analog of compatibility.

ω-compatibility: Let (X, ω^G) be a modular G-metric space and let $T_1, T_2 : X \to X$. The pair $\{T_1, T_2\}$ is *ω-compatible* if, for any sequence $\{x_n\}$ such that

$$T_1 x_n \to x \quad \text{and} \quad T_2 x_n \to x,$$

one has (for each $\lambda > 0$) asymptotic commutation in the modular G-metric, e.g.

$$\omega_\lambda^G(T_1 T_2 x_n,\ T_1 T_2 x_n,\ T_2 T_1 x_n) \to 0,$$

(or an equivalent symmetric variant).

Remark 1: switching (a, a, b) and (a, b, b) The paper uses a supporting proposition to move between expressions of the form $\omega_\lambda^G(a, a, b)$ and $\omega_\lambda^G(a, b, b)$, which is crucial when applying ω-compatibility alongside the inequality toolbox.

Main Results

Theorem 3.1. *Let (X_{ω^G}, ω^G) be a G-complete modular G-metric space and let $T_i : X_{\omega^G} \to X_{\omega^G}$ for $i = 1, 2, 3, 4$, be four self ω-compatible mappings with $T_1(X_{\omega^G}) \subseteq T_4(X_{\omega^G})$, $T_2(X_{\omega^G}) \subseteq T_3(X_{\omega^G})$ in which T_3, T_4 are continuous and that the pairs $\{T_1, T_3\}$ and $\{T_2, T_4\}$ are compatible so that there is a point $y_0 \in X_{\omega^G}$, $\lambda > 0$, such that $\omega^G_\lambda(y_1, y_1, y_0) < \infty$, for which the following condition is satisfied;*

$$\omega^G_\lambda(T_1x, T_1y, T_2z) \leq k\omega^G_\lambda(T_3x, T_3y, T_4z), \tag{3.1}$$

for each $x, y, z \in X_{\omega^G}$, with $k < 1$. Then T_i have a unique common fixed point in X_{ω^G} for $i = 1, 2, 3, 4$.

Start with an arbitrary $x_0 \in X$. Since $T_1(X) \subseteq T_4(X)$, there exists $x_1 \in X$ such that

$$T_1x_0 = T_4x_1.$$

Similarly, since $T_2(X) \subseteq T_3(X)$, we pick $x_2 \in X$ such that

$$T_2x_1 = T_3x_2.$$

Continuing inductively, for each $n \geq 0$ we choose points $x_{2n+1}, x_{2n+2} \in X$ satisfying

$$T_1x_{2n} = T_4x_{2n+1}, \qquad T_2x_{2n+1} = T_3x_{2n+2}.$$

Thus the four maps are interlaced. Define a sequence $\{y_n\}$ by

$$y_{2n} = T_1x_{2n} = T_4x_{2n+1}, \qquad y_{2n+1} = T_2x_{2n+1} = T_3x_{2n+2}.$$

This yields a single sequence $\{y_n\}$ in X. Applying the contractive hypothesis (3.1), we obtain

$$\omega^G_\lambda(T_1x_{2n}, T_1x_{2n}, T_2x_{2n+1}) \leq k\,\omega^G_\lambda(T_3x_{2n}, T_3x_{2n}, T_4x_{2n+1}).$$

By construction of the sequence $\{y_n\}$,

$$T_3x_{2n} = y_{2n-1}, \qquad T_4x_{2n+1} = y_{2n}.$$

Hence,

$$\omega^G_\lambda(y_{2n}, y_{2n}, y_{2n+1}) \leq k\,\omega^G_\lambda(y_{2n-1}, y_{2n-1}, y_{2n}).$$

This is equation (3.3), which yields the geometric recursion

$$d_n \leq k\,d_{n-1}$$

because

$$d_n := \omega^G_\lambda(y_{2n}, y_{2n}, y_{2n+1}),$$

which implies

$$d_{n-1} = \omega^G_\lambda(y_{2n-1}, y_{2n-1}, y_{2n}).$$

what happens after the recursion. The recursion $d_n \leq k\,d_{n-1}$ is the first "engine" step: it gives geometric decay of adjacent increments. In the paper, this is then strengthened using standard modular G-metric inequalities (derived from the rectangle inequality) to obtain a bound of the form

$$\omega^G_\lambda(y_{2n}, y_{2n}, y_{2n+1}) \leq \left(\frac{k}{2}\right)^{n-1} \omega^G_\lambda(y_0, y_0, y_1) \quad (n \geq 2),$$

which is strong enough to make the later telescoping estimate converge.

Step 2: show $\{y_n\}$ is modular G-Cauchy. Use the rectangle inequality repeatedly to chain y_{2n} to y_{2m} through consecutive indices:

$$\omega_\lambda^G(y_{2n}, y_{2n}, y_{2m}) \le \sum_{j=2n}^{2m-1} \omega_{\lambda/(m-n)}^G(y_j, y_{j+1}, y_{j+1}).$$

where the parameter splitting arises from repeated application of the rectangle inequality. Then bound each term $\omega_{\lambda/(m-n)}^G(y_j, y_{j+1}, y_{j+1})$ by the geometric decay from Step 1 above, turning the sum into a finite geometric series. This yields

$$\lim_{n,m\to\infty} \omega_\lambda^G(y_{2n}, y_{2n}, y_{2m}) = 0.$$

Finally, use the standard modular G-metric inequalities to upgrade this "even subsequence" estimate to the full 3-index Cauchy property:

$$\lim_{n,m,\ell\to\infty} \omega_\lambda^G(y_n, y_m, y_\ell) = 0.$$

Hence $\{y_n\}$ is modular G-Cauchy.

Step 3: completeness gives a limit u. By modular G-completeness, there exists $u \in X$ such that $y_n \to u$ in the modular G-sense. In particular, the interlaced subsequences converge to the same limit:

$$T_1x_{2n} \to u, \quad T_4x_{2n+1} \to u, \quad T_2x_{2n+1} \to u, \quad T_3x_{2n+2} \to u.$$

Step 4: prove u is a common fixed point. This step uses: (i) continuity of T_3 and T_4 on the orbit/at the limit, (ii) ω-compatibility of $\{T_1, T_3\}$ and $\{T_2, T_4\}$ (in the sense of modular G-metrics), and (iii) the contraction.

1. **Show $T_3u = u$.** Continuity implies $T_3(T_3x_{2n+2}) \to T_3u$ and also $T_3(T_1x_{2n}) \to T_3u$. ω-compatibility ensures the compositions $T_1T_3x_{2n}$ and $T_3T_1x_{2n}$ asymptotically coincide in ω_λ^G, hence share the same limit. Substitute appropriate (x, y, z) into the contraction and pass to the limit to get

   $$\omega_\lambda^G(T_3u, T_3u, u) \le k\, \omega_\lambda^G(T_3u, T_3u, u),$$

   which forces $\omega_\lambda^G(T_3u, T_3u, u) = 0$, hence $T_3u = u$.

2. **Show $T_4u = u$.** Similarly, continuity of T_4 and ω-compatibility of $\{T_2, T_4\}$ yield

   $$\omega_\lambda^G(u, u, T_4u) \le k\, \omega_\lambda^G(u, u, T_4u),$$

   hence $T_4u = u$.

3. **Show $T_1u = u$ and $T_2u = u$.** Once $T_3u = T_4u = u$, the contraction reduces to inequalities forcing $\omega_\lambda^G(T_1u, u, u) = 0$ and $\omega_\lambda^G(u, u, T_2u) = 0$. Thus $T_1u = u$ and $T_2u = u$.

Therefore u is a common fixed point.

Step 5: uniqueness. If u^* is another common fixed point, plug $(x, y, z) = (u, u, u^*)$ into the contraction:

$$\omega_\lambda^G(u, u, u^*) \le k\, \omega_\lambda^G(u, u, u^*).$$

Since $k < 1$, we get $\omega_\lambda^G(u, u, u^*) = 0$, hence $u = u^*$.

Explanation of $k/2$

We recall the standing definitions in the proof of Theorem 3.1:

$$y_{2n} = T_1 x_{2n} = T_4 x_{2n+1}, \qquad y_{2n-1} = T_2 x_{2n-1} = T_3 x_{2n}, \qquad y_{2n-2} = T_1 x_{2n-2} = T_4 x_{2n-1}.$$

In particular,

$$T_3 x_{2n} = y_{2n-1}, \qquad T_4 x_{2n-1} = y_{2n-2}, \qquad T_1 x_{2n} = y_{2n}, \qquad T_2 x_{2n-1} = y_{2n-1}.$$

The contractive hypothesis (3.1) says that for all $x, y, z \in X_{\omega_G}$ and all $\lambda > 0$,

$$\omega_\lambda^G(T_1 x, T_1 y, T_2 z) \leq k\, \omega_\lambda^G(T_3 x, T_3 y, T_4 z), \qquad 0 < k < 1. \tag{1}$$

Also Proposition 2.7(3) states that for all x, y and all $\lambda > 0$,

$$\omega_\lambda^G(x, y, y) \leq 2\, \omega_{\lambda/2}^G(y, x, x). \tag{2}$$

Apply (2) with $x = y_{2n-1}$ and $y = y_{2n}$. This gives

$$\omega_\lambda^G(y_{2n-1}, y_{2n-1}, y_{2n}) \leq 2\, \omega_{\lambda/2}^G(y_{2n}, y_{2n-1}, y_{2n-1}). \tag{3}$$

Using the identifications $y_{2n} = T_1 x_{2n}$ and $y_{2n-1} = T_2 x_{2n-1} = T_3 x_{2n}$, the right-hand side becomes

$$\omega_{\lambda/2}^G(y_{2n}, y_{2n-1}, y_{2n-1}) = \omega_{\lambda/2}^G(T_1 x_{2n}, T_2 x_{2n-1}, T_2 x_{2n-1}).$$

Now apply (1) with the substitution

$$x = x_{2n}, \qquad y = x_{2n}, \qquad z = x_{2n-1},$$

to obtain (at parameter $\lambda/2$)

$$\omega_{\lambda/2}^G(T_1 x_{2n}, T_1 x_{2n}, T_2 x_{2n-1}) \leq k\, \omega_{\lambda/2}^G(T_3 x_{2n}, T_3 x_{2n}, T_4 x_{2n-1}). \tag{4}$$

Since $T_3 x_{2n} = y_{2n-1}$ and $T_4 x_{2n-1} = y_{2n-2}$, (4) reads

$$\omega_{\lambda/2}^G(y_{2n}, y_{2n}, y_{2n-1}) \leq k\, \omega_{\lambda/2}^G(y_{2n-1}, y_{2n-1}, y_{2n-2}). \tag{5}$$

Combining (3) with (5) yields the explicit constant-tracking estimate

$$\omega_\lambda^G(y_{2n-1}, y_{2n-1}, y_{2n}) \leq 2k\, \omega_{\lambda/2}^G(y_{2n-1}, y_{2n-1}, y_{2n-2}). \tag{6}$$

The focus point to look is the following. Where the manuscript's $k/2$ can come from. In many fixed point arguments, authors re-parameterize the contraction constant after collecting auxiliary factors. (6) shows that the combination of Proposition 2.7(3) (which contributes the factor 2) and the contractive hypothesis (which contributes the factor k) produces a prefactor $2k$ when constants are tracked directly. If one wishes to present the subsequent iteration in geometric form

$$\omega_\lambda^G(y_{2n-1}, y_{2n-1}, y_{2n}) \leq q\, \omega_{\lambda/2}^G(y_{2n-1}, y_{2n-1}, y_{2n-2}) \qquad \text{with } 0 < q < 1,$$

then one may define $q := 2k$ and impose the strengthened requirement $2k < 1$ (equivalently $k < \frac{1}{2}$), or else one may restate the contractive assumption with a smaller constant so that after the factor 2 is introduced, the resulting q is still < 1. In particular, writing a coefficient like $\frac{k}{2}$ at that point corresponds to a normalization choice: it amounts to using a contraction constant that has already been adjusted to absorb the factor arising from Proposition 2.7(3). The fully expanded derivation under the displayed hypothesis (1) is (6).

Corollary

After the main theorem, many corollaries follow by specialization:

- Taking iterates T_i^m (powers) to obtain fixed points for iterated maps.
- Setting maps equal (e.g., $T_1 = T_2$, $T_3 = T_4$) to reduce four-map results to two-map or one-map versions.
- Setting $T_3 = T_4 = I$ (identity) to obtain modular-metric-like contractions.
- Replacing the basic contraction by a linear combination contraction with parameters (a, b, c, d) under constraints (e.g. $a+b+c+d < 1$), while keeping the same interlaced-orbit $\Rightarrow$ Cauchy $\Rightarrow$ limit $\Rightarrow$ fixed point mechanism.

Remark 2: Proof backbone Almost all variants preserve the same proof skeleton:

interlaced orbit $\Rightarrow$ geometric decay of increments $\Rightarrow$ Cauchy via rectangle inequality $\Rightarrow$ limit $\Rightarrow$ fixed point $\Rightarrow$ uniqueness.

Therefore, many future generalizations can be obtained by replacing only the contraction inequality while keeping the orbit/Cauchy mechanism intact—provided the proof dependencies are made correct and explicit.

Critical Analysis

A. Compatibility vs. ω-compatibility mismatch (major). The theorem statement in the paper uses the term "compatible" for the pairs $\{T_1, T_3\}$ and $\{T_2, T_4\}$, but the proof as written uses a modular-G version that requires limits like

$$\omega_\lambda^G(T_1T_3x_n,\ T_1T_3x_n,\ T_3T_1x_n) \to 0(\text{or symmetric variants})$$

That is exactly ω-compatibility in the modular G-metric sense (not classical metric/G-metric compatibility). "Compatibility" in metric/G-metric spaces and "ω-compatibility" in modular G-metric spaces are *different definitions* with different objects. If the proof uses ω-compatibility, it must be assumed (or derived). For extension work, the safe fix is:

- Restate the theorem assuming **ω-compatibility** explicitly, or
- Prove a bridge lemma that shows the stated compatibility implies ω-compatibility under additional structural assumptions.

B. λ-parameter handling and hidden monotonicity assumptions. Several bounding steps in the proof compare terms at different parameters (e.g. λ vs. $\lambda/2$, or $\lambda/(m-n)$ vs. other splits) when chaining rectangle inequalities. Unless the modular G-metric is known to be monotone in λ in the direction needed, such simplifications should be justified. Without a monotonicity property in λ, replacing $\omega_{\lambda/2}^G(\cdot)$ by $\omega_\lambda^G(\cdot)$ in a $\leq$ direction is not formally justified. For extension work, we should:

- add an explicit lemma/assumption describing the required monotonic behavior in λ, or

C. The "finite anchor" assumption. Since ω_λ^G may take value $+\infty$, the theorem includes an assumption of the form $\omega_\lambda^G(y_1, y_1, y_0) < \infty$ for some anchor. This is essential so the geometric bounds do not remain infinite. For extension work, it is cleaner to either assume $\omega_\lambda^G < \infty$ everywhere on X or explicitly specify how the initial point is chosen to guarantee finiteness of the starting quantity.

D. Where continuity is used (opportunity for weakening). Continuity of T_3 and T_4 is used to pass limits along the constructed orbit and identify limits of compositions before applying ω-compatibility (e.g., pass from $y_n \to u$ to $T_3 y_n \to T_3 u$ and $T_4 y_n \to T_4 u$ on the *specific orbit*.) This suggests that global continuity may be stronger than necessary. Often, the argument only needs:

- sequential continuity at the limit point u, or
- orbital continuity along $\{y_n\}$,

rather than global continuity.

E. Example not fully self-contained. The illustrative example definea explicit T_i on a set like $\mathbb{R}_+ \cup \{\infty\}$, but the modular G-metric ω_λ^G is sometimes referenced externally. For an extension paper, a reusable example should fully specify:

1. Define ω_λ^G on the given set and verify the modular G-metric axioms.
2. Verify ω-compatibility (not just classical compatibility) for the maps.
3. Verify the contraction inequality directly.

Extension-Ready Checklist

Before using these results as a base for an extension, we believe a safe workflow is:

1. **Restate the core theorem** with hypotheses matching the proof (especially ω-compatibility).
2. **Make λ-behavior explicit:** either assume monotonicity or rewrite estimates to avoid it.
3. **Isolate a reusable orbit/Cauchy lemma:** (interlaced orbit + geometric decay + rectangle inequality $\Rightarrow$ Cauchy).
4. **Then swap in new contractions:** e.g., control functions (Geraghty/Meir–Keeler/Reich/Cirić-type), weakening continuity, adding order/graph structures, etc.
5. **Provide at least one fully self-contained example** in the modular G-metric setting.